

Pitch Autopilot Controller

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Nomenclature

V_T	True Airspeed
α	Angle of Attack
θ	Pitch Angle
q	Pitch Rate
δ_e	Elevator Deflection
x	State Vector
u	Control Input
y	Output Vector
A	System Matrix
B	Input Matrix
C	Output Matrix
D	Feedthrough Matrix

Abstract

This report presents the development of a pitch autopilot controller for a longitudinal aircraft model. A linearized state-space representation of the aircraft dynamics is used to analyze the open-loop behavior and design a feedback control system. The controller is evaluated through simulation to assess stability, transient response, and command-tracking performance. The results demonstrate the effectiveness of the autopilot in improving aircraft pitch regulation and overall dynamic response.

1. Governing equations for longitudinal dynamics

The linearized longitudinal dynamics of the aircraft about a steady flight condition are given by:

$$\begin{aligned}\dot{V}_T &= -0.03V_T - 32.1\theta \\ \dot{\theta} &= 56.7q \\ \dot{\alpha} &= -0.00025V_T - 0.313\alpha + 56.7q + 0.232\delta_e \\ \dot{q} &= -0.0139\alpha - 0.426q + 0.0203\delta_e\end{aligned}$$

2. State Space Representation

$$\dot{\vec{x}} = A\vec{x} + B\vec{u}$$

$$\begin{bmatrix} \dot{V}_T \\ \dot{\theta} \\ \dot{\alpha} \\ \dot{q} \end{bmatrix} = \begin{bmatrix} -0.03 & -32.1 & 0 & 0 \\ 0 & 0 & 0 & 56.7 \\ -0.00025 & 0 & -0.313 & 56.7 \\ 0 & 0 & -0.0139 & -0.426 \end{bmatrix} \begin{bmatrix} V_T \\ \theta \\ \alpha \\ q \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0.232 \\ 0.0203 \end{bmatrix} \delta_e$$

$$\vec{y} = C\vec{x} + D\vec{u}$$

$$y = \begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} V_T \\ \theta \\ \alpha \\ q \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} \delta_e$$

3. Preliminary System Analysis

The eigenvalues of the system matrix, computed in MATLAB using the `eig(A)` command, are $\lambda_{1,2} = -0.3722 \pm 0.8831i$ and $\lambda_{3,4} = -0.0123 \pm 0.0821i$. Since the real part of each eigenvalue is negative, i.e., $\text{Re}(\lambda_i) < 0$ for all i , the open-loop system is asymptotically stable. The controllability matrix was obtained using the `ctrb(A,B)` command, and its rank was verified with `rank(Co)`, confirming that the system is fully controllable. Similarly, the observability matrix was computed using `obsv(A,C)`, and its rank was verified using `rank(Ob)`. Since the observability matrix has full rank, the system is also fully observable.

4. Simulink Implementation

Before implementing the complete autopilot control architecture, the aircraft state-space model is first constructed in Simulink. Although a built-in State-Space block could be used, implementing the model explicitly allows the intermediate state signals to be accessed and monitored more easily during analysis and controller development.

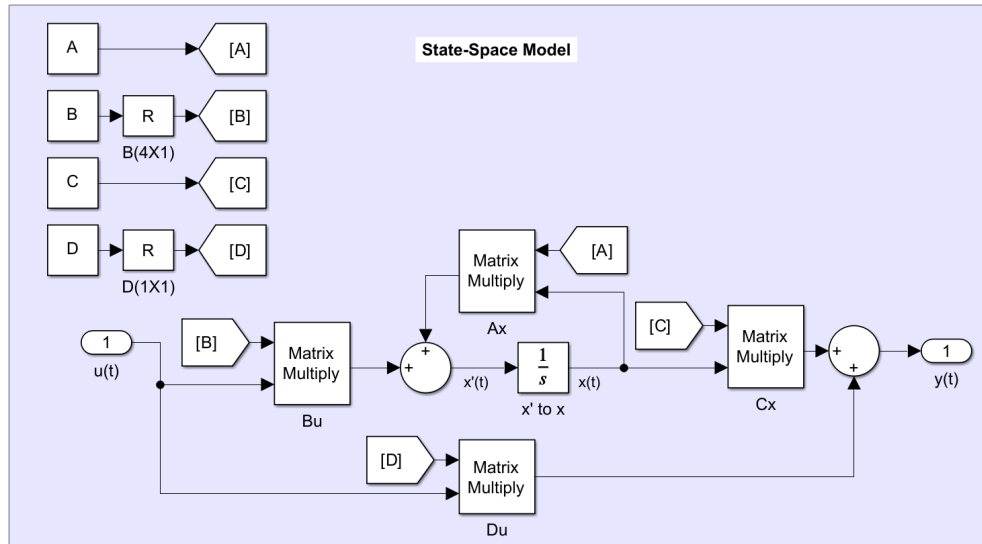


Figure 1: State-Space Model Block

Once the state-space model has been implemented, the complete autopilot control architecture can be developed, as illustrated below.

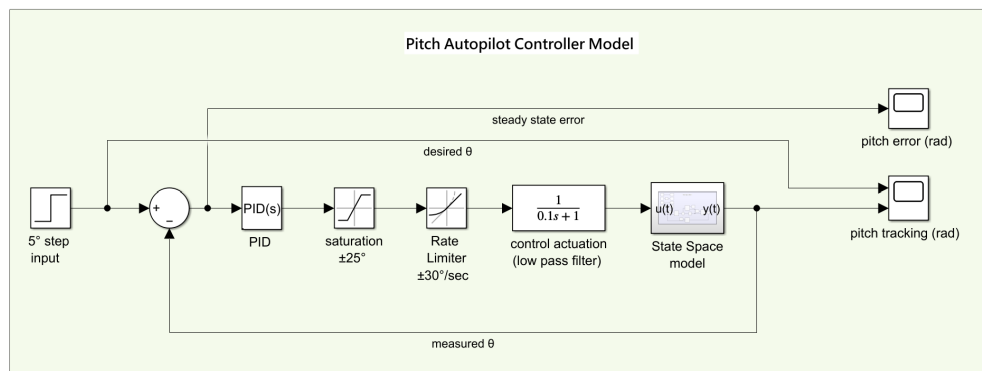


Figure 2: Preliminary autopilot Model

The closed-loop pitch control system begins with a 5° step input, representing the desired pitch angle, θ , commanded by either the pilot or the autopilot. This reference signal enters a summing junction, where it is compared with the measured aircraft pitch angle to generate the tracking error. The PID controller processes this error and computes the appropriate control input required to minimize the tracking error, thereby driving the aircraft toward the commanded pitch angle.

Before reaching the aircraft dynamics, the controller output passes through several blocks that model the physical limitations of the flight control system. A saturation block limits the maximum control surface deflection to $\pm 25^\circ$, reflecting the mechanical constraints of the elevator. A rate limiter further restricts the actuator speed to $\pm 30^\circ/\text{s}$, preventing unrealistically rapid control surface movements. The control actuation dynamics are then represented by a first-order low-pass transfer function,

$$\frac{1}{0.1s + 1},$$

which models the finite response time and inherent lag of the actuator.

The conditioned control signal is subsequently applied to the state-space model of the aircraft, which represents the longitudinal flight dynamics. Using the actuator input, $u(t)$, the model computes the corresponding aircraft states and outputs, $y(t)$, including the pitch angle, θ . The measured pitch angle is fed back to the summing junction, completing the closed-loop feedback system.

System performance is evaluated using two scope displays. The first records the pitch tracking error as a function of time, providing insight into the transient and steady-state response. The second compares the commanded and actual pitch angles, allowing the tracking accuracy, overshoot, settling time, and overall closed-loop stability to be visually assessed.

5. PID Tuning

There are several established methods for tuning a PID controller. Rather than using the built-in PID Tuner provided by Simulink, a custom tuning dashboard was developed to streamline the tuning process. This interface allows the proportional, integral, and derivative gains to be adjusted interactively while providing immediate visual feedback of the system response, eliminating the need to repeatedly open and navigate multiple windows during controller tuning. The primary limitation of this approach is the reduced display fidelity of the Dashboard Scope compared to the standard Simulink Scope.

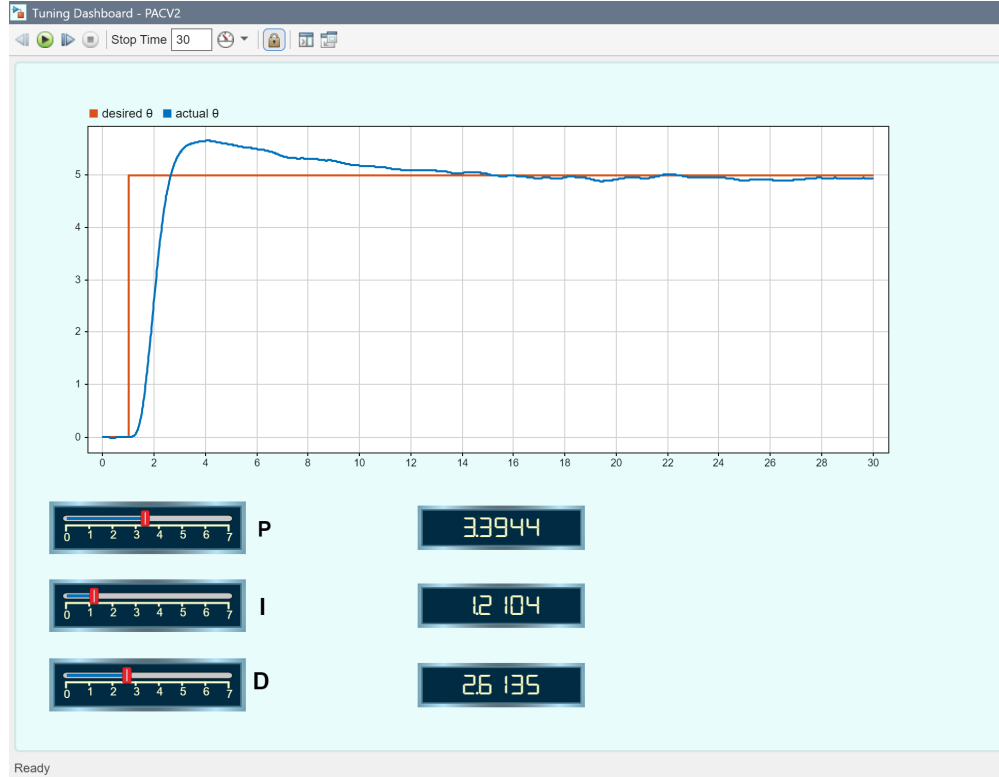


Figure 3: Tuning dashboard in simulink

Once the dashboard has been connected to the appropriate signals, the standard PID tuning procedure can be applied. The proportional gain, K_p , is increased first until the system achieves a fast response while maintaining acceptable stability. Next, the integral gain, K_i , is introduced to eliminate the steady-state error, taking care to avoid excessive overshoot or oscillations. Finally, the derivative gain, K_d , is adjusted to improve damping, reduce overshoot, and shorten the settling time, resulting in a well-balanced closed-loop response.

6. Results of Preliminary Model

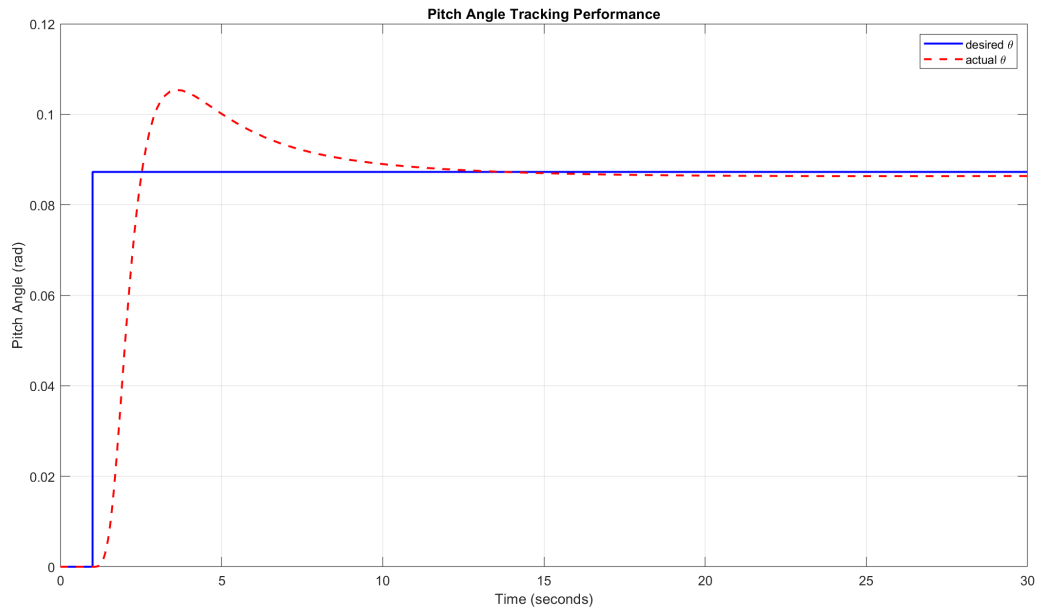


Figure 4: Pitch tracking

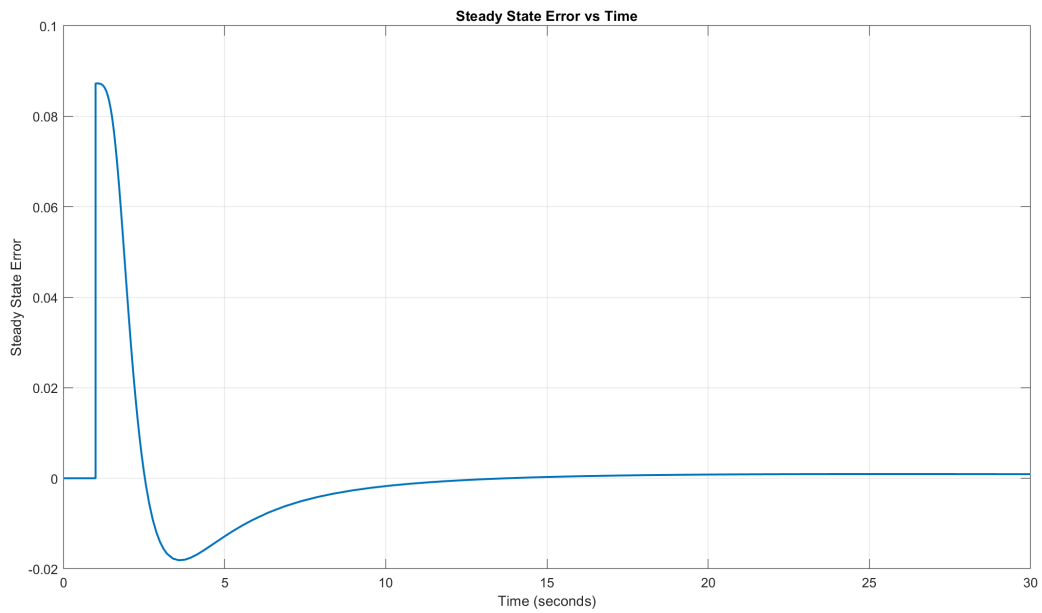


Figure 5: Steady State Error

The transient response of the closed-loop system begins at $t = 1$ s, when a 5° step command (approximately 0.087 rad) is applied to the reference input. As shown in the Pitch Angle Tracking Performance plot, the actual pitch angle (red dashed line) responds with a slight delay due to the actuator dynamics before rapidly converging toward the commanded pitch angle (blue solid line). The response exhibits a modest overshoot, reaching a peak value of approximately 0.105 rad at around $t = 3.5$ s. Following this peak, the system displays a lightly underdamped response, with the oscillations decaying smoothly until the pitch angle settles at the commanded value.

The Steady-State Error plot provides a complementary view of the system performance by illustrating the instantaneous difference between the commanded and measured pitch angles. Immediately after the step input is applied at $t = 1$ s, the tracking error reaches its maximum value of approximately 0.087 rad, as the aircraft has not yet begun to respond. As the pitch angle increases toward the reference, the error decreases rapidly, crosses zero during the overshoot, and reaches a minimum of approximately -0.018 rad. The PID controller subsequently attenuates the remaining error, driving it to zero by approximately $t = 15$ s, thereby demonstrating excellent steady-state tracking performance with negligible residual error.

7. Improvements on controller

Overall, the current controller demonstrates satisfactory tracking performance under ideal simulation conditions. However, the system is not yet fully representative of a real aircraft, where sensor noise, external disturbances, actuator uncertainties, and unmodeled dynamics are inevitably present. Consequently, the next stage of development is to investigate the robustness of the controller by introducing measurement noise and disturbance inputs into the simulation. This will allow the controller's ability to reject disturbances and maintain accurate pitch tracking under more realistic operating conditions to be evaluated.

In addition, the controller has primarily been tuned and validated using a single 5° step input. Although the response under this operating condition is satisfactory, there is no guarantee that the same controller will exhibit comparable performance for larger or smaller step commands, rapidly varying reference inputs, or other flight conditions. Future work will therefore focus on evaluating the controller over a broader range of reference inputs and operating scenarios, with the objective of developing a controller that maintains consistent performance, stability, and robustness across the aircraft's operating envelope rather than being optimized for a single test case.

8. Controller Design Version Two

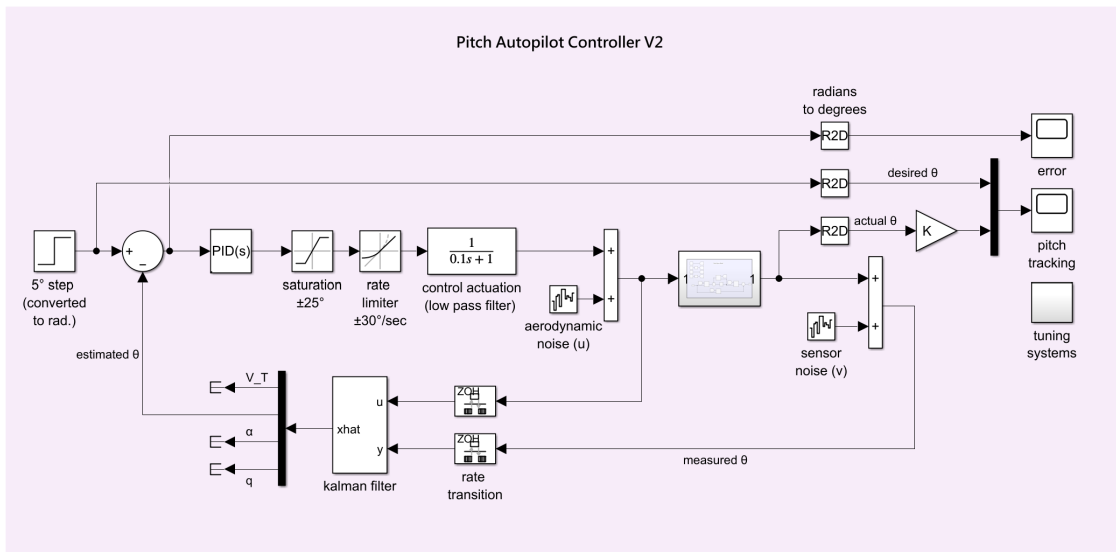


Figure 6: Improved controller design

The controller design was enhanced to improve both the realism and usability of the simulation. Two key additions were incorporated into the existing model: a Kalman filter for state estimation and simulated sensor and aerodynamic noise to represent real-world operating conditions. Additionally, radians-to-degrees conversions were incorporated where appropriate to provide a more intuitive interface for controller tuning and analysis while preserving internal calculations in radians.

9. Results for Controller Model 2

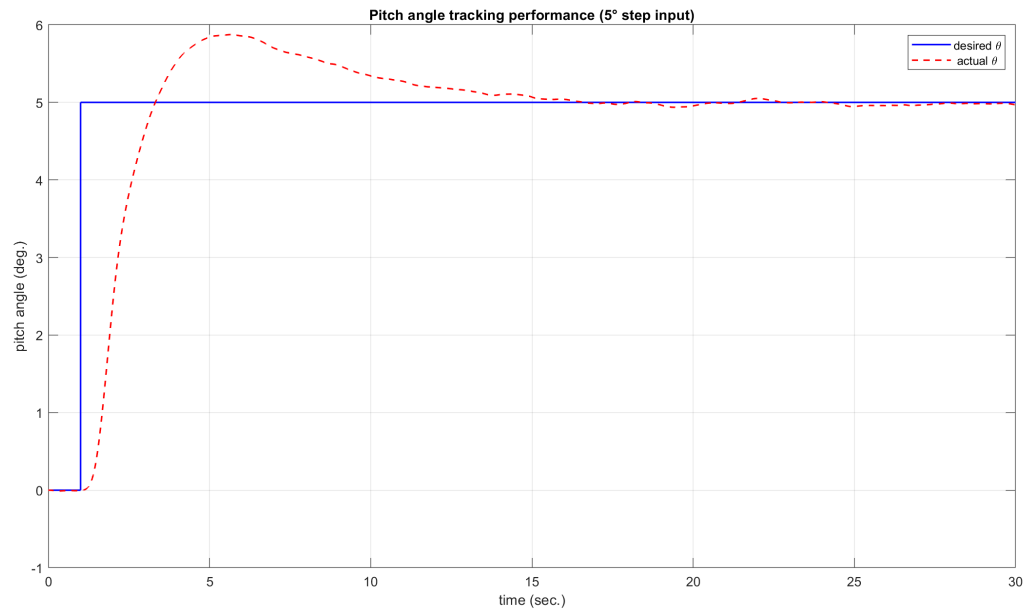


Figure 7: 5 degree step input (with integrator windup)

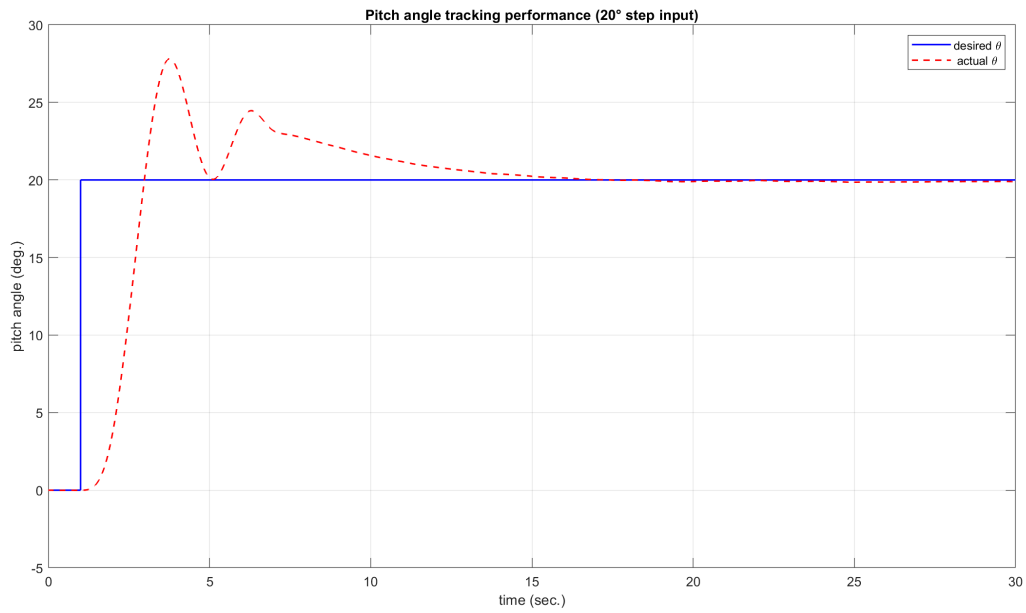


Figure 8: 20 degree step input (with integrator windup)

Up to this stage, controller performance metrics have not been a primary focus, as the PID controller was still under development. Initial tuning was performed by approximation using a single 5° step input to establish satisfactory closed-loop behavior. With the controller architecture now approaching its final form, a more rigorous evaluation of the transient response is required. To assess its performance over a wider operating range, the reference input was increased to a 20° step and the resulting response was compared with that obtained for the original 5° command.

This evaluation revealed that the controller does not maintain consistent performance as the step magnitude increases. In particular, the overshoot becomes more pronounced for larger reference inputs. Furthermore, despite extensive manual tuning of the PID gains, the overshoot could not be significantly reduced without adversely affecting other aspects of the response, indicating a limitation of the current fixed-gain PID controller.

A likely cause of this behavior is integrator windup. When the controller output reaches the actuator limits, the integral term can continue accumulating error even though the actuator is saturated. Once the actuator leaves saturation, the stored integral action produces an excessively large control effort, resulting in increased overshoot and a longer settling time. To mitigate this effect, an anti-windup mechanism will be incorporated into the PID controller. Anti-windup prevents the integral term from accumulating excessively during actuator saturation, thereby improving the transient response and maintaining consistent performance over a wider range of reference inputs.

10. Final Controller Design and FlightGear Implementation

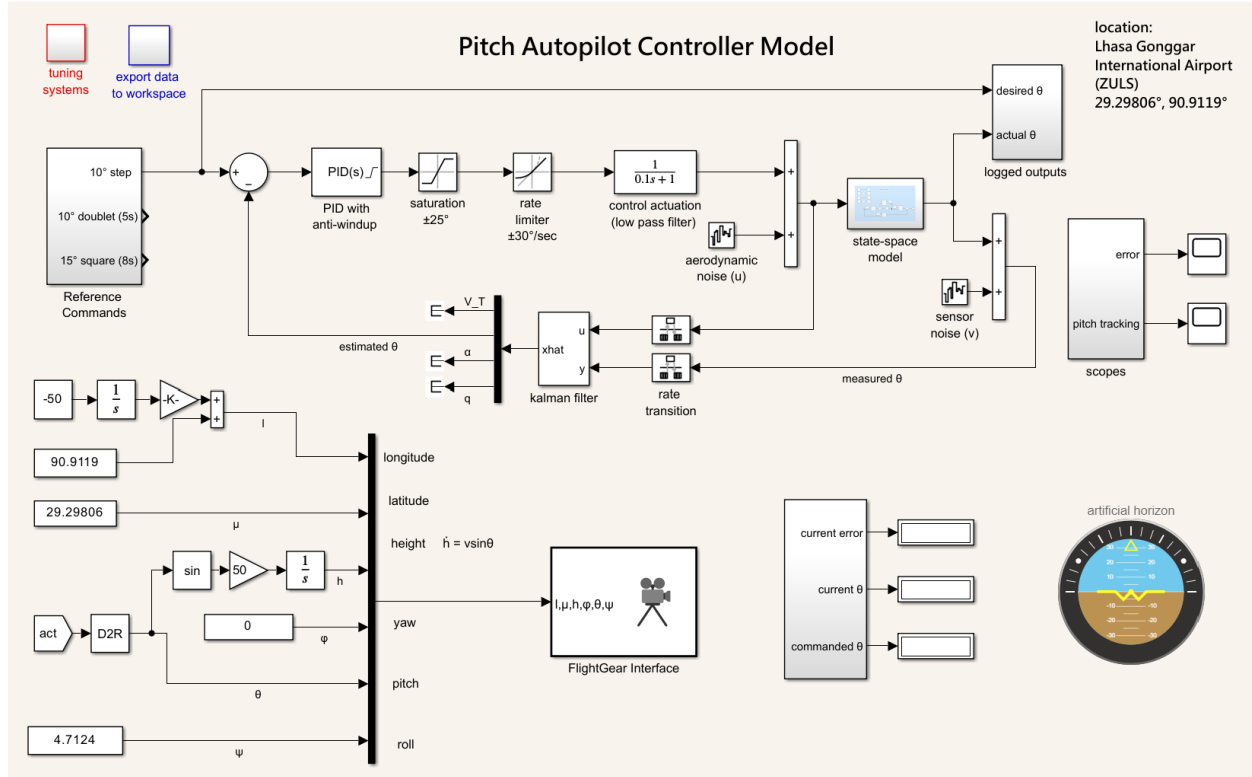


Figure 9: Final controller design

The final Pitch Autopilot Model introduces enhanced reference command generation, improved visualization, and a cleaner overall layout compared to the previous version. The original fixed 5° step input has been replaced with a *Reference Commands* subsystem that provides a 10° step, a 10° doublet (5 s), and a 15° square wave (8 s), allowing the controller to be evaluated under a variety of standard test inputs.

For visualization, the model incorporates a FlightGear interface that generates the navigation and attitude states required for real-time flight visualization, including latitude, longitude, altitude, yaw, pitch, and roll. An artificial horizon and numerical displays for the commanded pitch angle, measured pitch angle, and tracking error provide real-time feedback on controller performance.

Finally, long signal connections have been replaced with Simulink *Goto* and *From* blocks, improving readability, reducing visual clutter, and creating a more organized signal flow throughout the model.

NOTE: Final results are on website